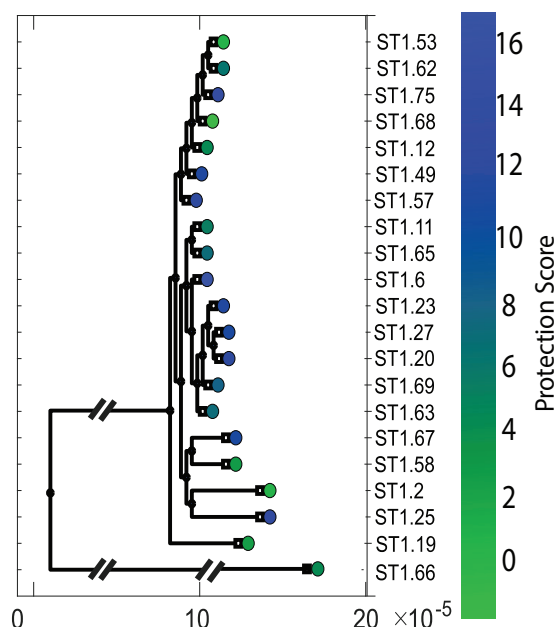
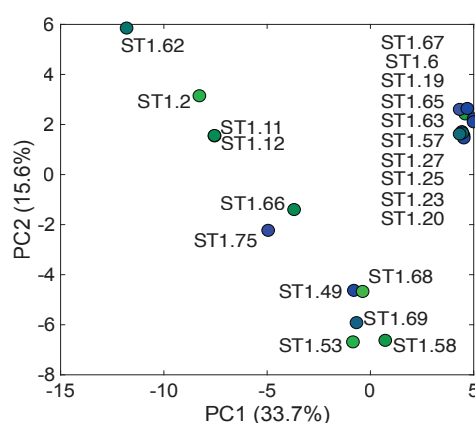


A



B



C

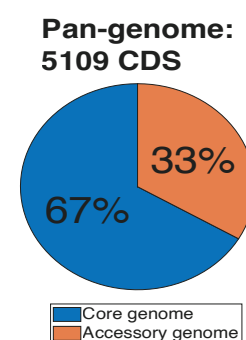


Figure 4: Genomic analysis of ST1 strains does not reveal a single determinant of protection. (A) Phylogenetic tree of 21 ST1 clinical isolates based on core genome single-nucleotide substitutions. Branch tips are colored by protection score (color scale, right). No distinct phylogenetic clustering separates protective from non-protective strains. (B) Principal component analysis (PCA) of the accessory genome presence/absence matrix for all 21 ST1 strains. PC1 explains 33.7% and PC2 explains 15.6% of the variance. Strains are colored by protection score as in (A). No clear separation between protective and non-protective strains is observed. (C) Pan-genome composition of the 21 ST1 strains, comprising 5,109 total coding sequences (CDS). The core genome (shared across all strains, blue) accounts for 67% of CDS, and the accessory genome (present in a subset of strains, orange) accounts for 33%.